Introduction to Fourier Neural Operators

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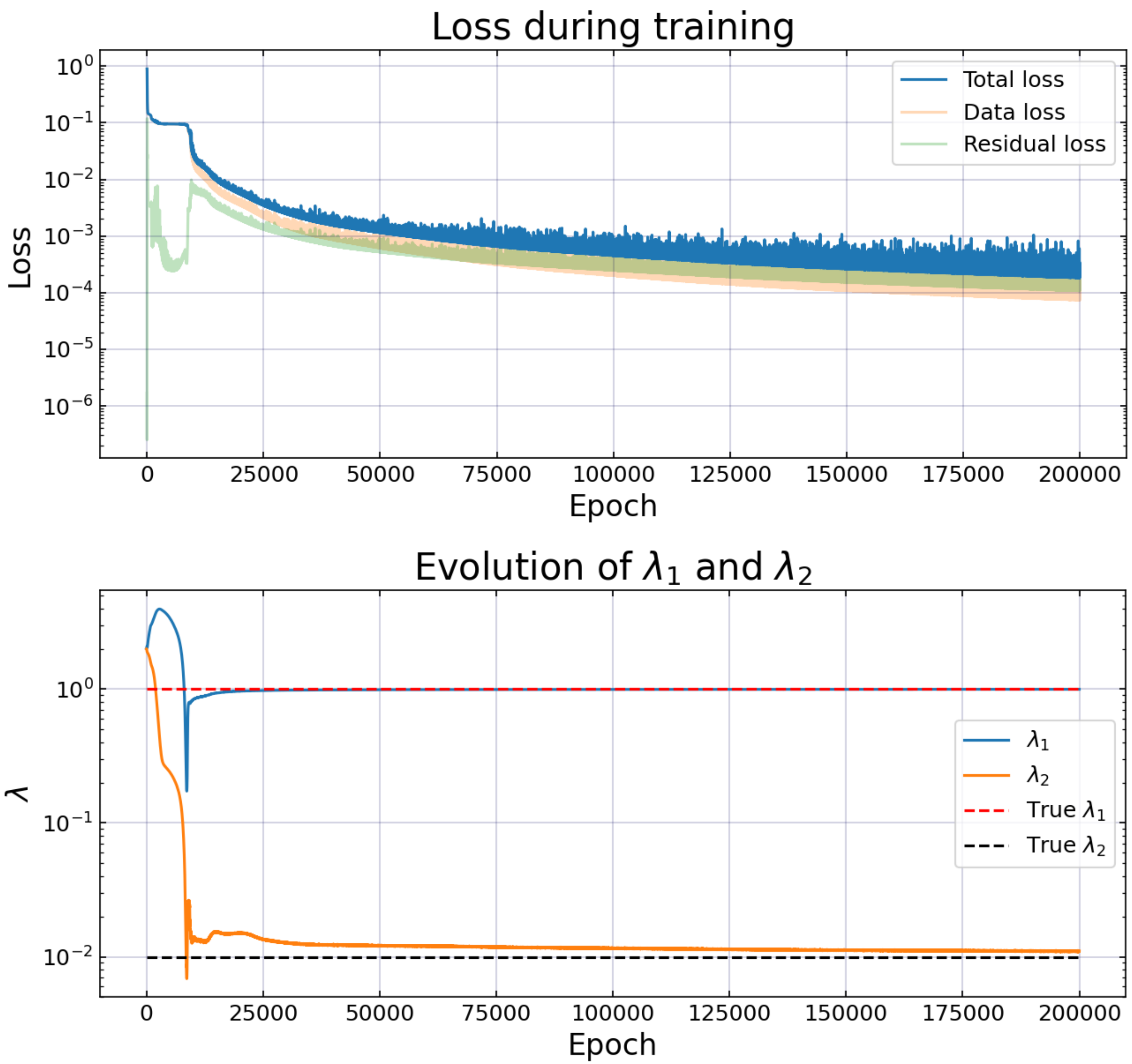
Data driven discovery of PDEs

Navier-Stokes Equation:

$$u_t + \lambda_1\left(uu_x + vu_y\right) = -p_x\lambda_2\left(u_{xx} + u_{yy}\right)$$
$$v_t + \lambda_1\left(uv_x + vv_y\right) = -p_y\lambda_2\left(v_{xx} + v_{yy}\right)$$

Estimate true values for λ_1 and λ_2 using PINN:

- 5000 data points
- 200000 epochs using Adam optimizer
- 50000 iterations using L-BFGS optimizer



Loss during training
Total loss
Data loss
Residual loss
Loss
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
25000
50000
75000
100000
125000
150000
175000
200000
Epoch
Evolution of λ_1 and λ_2
λ_1
λ_2
True λ_1
True λ_2
λ
10^0
10^{-1}
10^{-2}
0
25000
50000
75000
100000
125000
150000
175000
200000
Epoch

Results

> 9 hours of training time

Parameters	True value	Predicted value	Relative error
λ_1	1	0.99713	0.29%
λ_2	0.01	0.01101	10.07%

What if we want to predict a solution to the Navier-Stokes equation using the same parameters but different initial/boundary conditions?

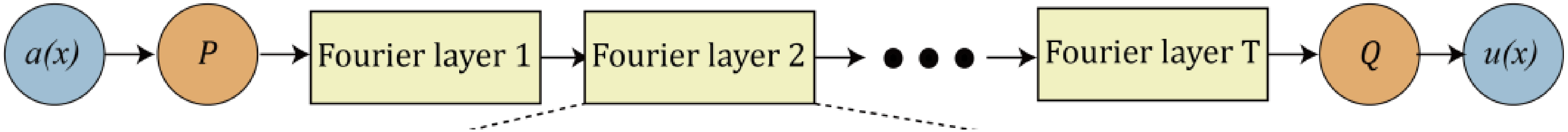
Fourier Neural Operators (FNOs)

- Train a solution operator for a whole family of PDEs
- Model learns $G: \mathcal{A} \times \Theta \rightarrow \mathcal{U}$
- Combines linear global integral operators and non-linear local activation functions

- Assuming translational invariance
- Model integral operator as convolution → work in Fourier space
- Use Fast Fourier Transform (FFT): $\mathcal{O}(n^2) \rightarrow \mathcal{O}(n \log n)$

High level architecture

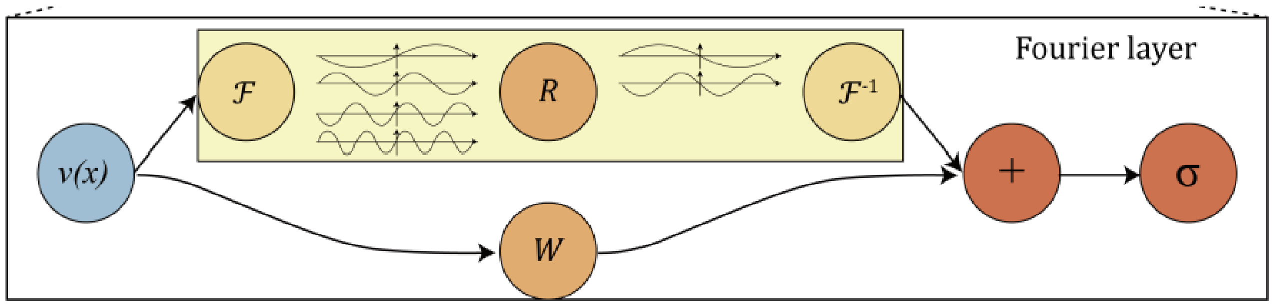
1. Start from input a ϵ $\mathcal{A}$
2. Lift to higher dimension channel space by neural network P
3. Apply multiple layers of integral operators and activation functions
4. Project back to target dimension by neural network Q
5. Output u ϵ $\mathcal{U}$



Fourier layers

Iterative update scheme:

$$v_{t+1}(x) \coloneqq \sigma\big((\mathcal{K}v_t)(x) + Wv_t(x)\big)$$



Global path (Fourier integral operator $\mathcal{K}$):

- $(\mathcal{K}v_t)(x) = \mathcal{F}^{-1}\big(R \cdot (\mathcal{F}v_t)\big)$
- Truncate Fourier modes to k_{max} to suppress noise

Improvements on PDE solutions

Burgers' Equation:

- 30% lower error rate

Darcy Flow:

- 60% lower error rate

Navier-Stokes:

- 30% lower error rate
- $< 1\%$ error with Reynolds number 1000
- 8% error with Reynolds number 10000

Application to Navier-Stokes

Estimate the solution to Navier-Stokes equation given sparse data in spatial and time domains

- Fix $\lambda_1 = 1$, parameterise $\lambda_2 = \nu$
- Spatial resolution 64×64
- Time domain [0,10] in 20 steps

Config	**Parameters**	**Time per epoch**	$\nu = 1\text{e}{-3}$ $T = 50$ $N = 1000$	$\nu = 1\text{e}{-4}$ $T = 30$ $N = 1000$	$\nu = 1\text{e}{-4}$ $T = 30$ $N = 10000$	$\nu = 1\text{e}{-5}$ $T = 20$ $N = 1000$
FNO-3D	$6,558,537$	$38.99s$	**0.0086**	0.1918	**0.0820**	0.1893
FNO-2D	$414,517$	$127.80s$	0.0128	**0.1559**	0.0834	**0.1556**
U-Net	$24,950,491$	$48.67s$	0.0245	0.2051	0.1190	0.1982
TF-Net	$7,451,724$	$47.21s$	0.0225	0.2253	0.1168	0.2268
ResNet	$266,641$	$78.47s$	0.0701	0.2871	0.2311	0.2753

Use cases

Spectral analysis

- Truncation of Fourier modes does not limit number of modes to k_{max}
- Activation function σ and decoder Q can recover high frequency modes
- For $\nu = 10^{-3}$, truncating analytic solution to 20 modes gives 2% error
- FNO recovers solution with $\leq 1\%$ error when only using 12 modes

Zero-shot super-resolution

- FNO can identify high frequency patterns
- Train operator on low resolution data and evaluate at a higher resolution
- Trained FNO-3D model on $64 \times 64 \times 20$ resolution data
- Able to transfer to $256 \times 256 \times 80$ resolution

Use cases

Non-periodic boundary conditions

- Traditional Fourier methods only work with periodic boundary conditions
- Linear transform W in the local path of Fourier layers enforces non-periodicity

Bayesian Inverse Problems

- Inferring parameters from observed data while quantifying uncertainty
- 2.5s → 0.005s per evaluation using FNO compared to pseudo-spectral methods
- Total training time: 18hrs → 2.5mins
- FNO can predict solutions for different initial conditions and observations

Thank for listening